

JEE MAIN REVISION MASTER: BINOMIAL THEOREM 2026 1 QUESTIONS 1.67% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Binomial Theorem - Core Application	9 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Binomial Theorem - Core Application (High Yield — 9 PYQs)

Core Principle: Key problem pattern in Kcet Binomial Theorem.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The value at $x=2$ for $\frac{x^3+3x^2+3x+1}{x^4+4x^3+6x^2+4x+1}$..." [Ans: C]
- [2025 Shift] Q#2 (MCQ): "If the number of terms in the binomial expansion of $(2x+3)^{3n}$ is 22, then the value of n is..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "The value of $^{49}C_3 + ^{48}C_3 + ^{47}C_3 + ^{46}C_3 + ^{45}C_3 + ^{45}C_4$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The value at $x=2$ for $\frac{x^3+3x^2+3x+1}{x^4+4x^3+6x^2+4x+1}$

- (A) 3
- (B) $(25 / 61)$
- (C) $(1 / 3)$ **[Correct]**
- (D) $(19 / 73)$

Correct Answer: C

Solution Analysis: The given expression is $\frac{x^3+3x^2+3x+1}{x^4+4x^3+6x^2+4x+1}$. We can recognize both numerator and denominator using identities. $x^3+3x^2+3x+1=(x+1)^3$ and $x^4+4x^3+6x^2+4x+1=(x+1)^4$. So the expression becomes $\frac{(x+1)^3}{(x+1)^4}$. Cancel the common factor $(x+1)^3$: $\frac{(x+1)^3}{(x+1)^4} = \frac{1}{x+1}$. Now put $x=2$: $\frac{1}{2+1} = \frac{1}{3}$. So the correct option is $(1 / 3)$ Option C.